ATIKUL HOQUE

Ph.D. Candidate, Geography and Environmental Studies | New Mexico State University | Expected May 2027
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Google Scholar | GitHub | LinkedIn | Website

RESEARCH PROFILE

I develop deep learning and multi-sensor remote sensing methods to quantify vegetation change and drought in water-limited ecosystems. My dissertation couples UAS hyperspectral imagery with Sentinel-1 SAR and Sentinel-2 time series to map woody plant encroachment from individual shrubs to landscape fractional cover at the Jornada Experimental Range. I also build open, operational products, including a weekly 5 km ecological drought index for the conterminous United States distributed through Google Earth Engine.

Dryland remote sensing · Woody plant encroachment · Deep learning and GeoAI · SAR and optical data fusion · Fractional vegetation cover · Vegetation phenology · Drought monitoring · UAS remote sensing

EDUCATION

Ph.D., Geography and Environmental Studies 2022 – May 2027 (expected)
New Mexico State University, Las Cruces, NM

- Dissertation: *Mapping and Monitoring of Shrub-Encroached Vegetation in Drylands Using Remote Sensing*
- Advisor: Dr. Michaela Buenemann. Committee: Dr. Niall P. Hanan, Dr. Chao Fan, Dr. Liping Yang (University of New Mexico)
- Admitted to candidacy; dissertation proposal defended September 2026. GPA 3.98/4.00

M.Sc., Natural Resource Management 2020 – 2022
Faculty of Geo-Information Science and Earth Observation (ITC), University of Twente, Enschede, The Netherlands

- Thesis: *Variability of Wheat Phenology from Sentinel-1 and -2 Time Series: A Case Study for Brandenburg, Germany*
- GPA 7.95/10. Orange Knowledge Programme and ITC Excellence Scholar

B.Sc., Geography and Environmental Studies 2015 – 2019
University of Chittagong, Chattogram, Bangladesh

- GPA 3.70/4.00

PUBLICATIONS

Peer-Reviewed Journal Articles

4. **Hoque, A.**, Mahmud, A., Rashid, K. J., & Biswas, R. N. (2026). Assessing spatial variability of Boro rice (*Oryza sativa* L.) phenology across different climatic zones in Bangladesh using Sentinel-1 and -2 time series data. *Agricultural Research*, 1–24. https://doi.org/10.1007/s40003-025-00921-z
3. Mahmud, A., **Hoque, A.**, Tuure, J., Alakukku, L., Mäkelä, P. S., Wachiye, S., … & Heiskanen, J. (2026). Integrating field camera imagery for monitoring maize phenology, biophysical traits, and agroclimatic factors in smallholder farms. *Computers and Electronics in Agriculture*, 240, 111161. https://doi.org/10.1016/j.compag.2025.111161
2. Paul, A., **Hoque, A.**, Mahmud, A., Esha, T. A., Rashid, K. J., Paul, B., … & Nath, T. K. (2024). Water, sanitation, and hygiene conditions and health outcomes: A comparative study between Rohingya refugees and local people in southeastern Bangladesh. *The Dhaka University Journal of Earth and Environmental Sciences*, 13(1), 47–58. https://doi.org/10.3329/dujees.v13i1.77575
1. Rashid, K. J., **Hoque, A.**, Esha, T. A., Rahman, M. A., & Paul, A. (2021). Spatiotemporal changes of vegetation and land surface temperature in the refugee camps and surrounding areas of Bangladesh after the Rohingya influx from Myanmar. *Environment, Development and Sustainability*, 23, 3562–3577. https://doi.org/10.1007/s10668-020-00733-x

Manuscripts in Preparation

Hoque, A., Buenemann, M., Fan, C., Hanan, N. P., & Yang, L. From pixels to landscapes: Advances in remote sensing of shrub encroachment in global drylands.

Hoque, A., Buenemann, M., Fan, C., Hanan, N. P., & Yang, L. A hybrid CNN-MLP approach for classifying photosynthetic vegetation, non-photosynthetic vegetation, and soil in mesquite-encroached rangelands from UAS imagery.

Hoque, A., Buenemann, M., Fan, C., Hanan, N. P., & Yang, L. Sentinel-1 and Sentinel-2 fusion with an attention-based CNN for fractional shrub cover mapping in drylands.

Hoque, A., et al. EcoDRI: A weekly ecological drought index for the conterminous United States, 2001–2025. Target journal: *Scientific Data*.

Thesis

Hoque, A. (2022). *Variability of wheat phenology from Sentinel-1 and -2 time series: A case study for Brandenburg, Germany* [M.Sc. thesis]. University of Twente, Enschede, The Netherlands.

RESEARCH EXPERIENCE

Doctoral Researcher, Spatial Applications Research Center (SpARC) 2022 – Present
New Mexico State University. Graduate Research Assistant (2022 – 2025). Supervisor: Dr. Michaela Buenemann

- Developed a hybrid CNN-MLP classifier that separates photosynthetic vegetation, non-photosynthetic vegetation, and bare soil in UAS hyperspectral imagery of mesquite-encroached rangelands, validated across 13 sites at the Jornada Experimental Range.
- Designed an attention-based CNN that fuses Sentinel-1 SAR and Sentinel-2 multispectral data to retrieve fractional shrub cover at landscape scale, using UAS-derived classifications as reference data.
- Conducted a systematic review of 174 studies (234 sensor and method configurations) on remote sensing of woody vegetation in global drylands.
- Currently testing climate-informed acquisition timing for shrub cover retrieval from Sentinel-2 time series (dissertation Chapter 4).
- Planned and led field campaigns as an FAA Part 107 remote pilot, collecting UAS imagery and field spectroradiometer measurements for radiometric calibration and validation.
- Supported collaborative projects on optical and radar land cover change mapping in north-central Panama and GeoAI-based prediction of acequia distribution in New Mexico.

Independent Research: Ecological Drought Index (EcoDRI) 2025 – Present
New Mexico State University

- Built a weekly 5 km drought index for the conterminous United States (2001–2025) that blends vegetation, thermal, and soil moisture condition indices with the 90-day SPI, weighted by Köppen–Geiger climate zone.
- Recalibrated drought categories against the U.S. Drought Monitor and evaluated the index with USDA NASS crop yields and RAP/RCMAP rangeland vegetation data across water-limited biomes.
- Released an interactive Google Earth Engine application and open-source processing code.

Research Intern, German Remote Sensing Data Center (DFD) 2022
German Aerospace Center (DLR), Oberpfaffenhofen (Munich), Germany. Funded by the Erasmus Mobility Exchange Fund

- Built a Google Earth Engine workflow to extract Sentinel-2 vegetation index profiles across winter wheat phenological stages.
- Analyzed spatial and interannual variability of winter wheat phenology from SAR and optical time series combined with gridded weather and soil data.
- Prepared field parcel data in Python and QGIS for crop type classification pipelines.

Teaching and Research Assistant 2019 – 2020
Department of Geography and Environmental Studies, University of Chittagong, Bangladesh

- Processed and analyzed geospatial and climate data for departmental research projects.

OPEN DATA AND SOFTWARE

- **EcoDRI-CONUS**: Ecological drought index for the conterminous United States, with a Google Earth Engine viewer and modular export scripts. github.com/Atikul-itc/EcoDRI-CONUS

CONFERENCE PRESENTATIONS

Oral Presentations

Hoque, A., & Buenemann, M. (2026). Hybrid deep learning for UAS-based shrubland mapping: Optimizing spatial resolution and spectral features. *American Association of Geographers (AAG) Annual Meeting*, San Francisco, CA.

Hoque, A., & Buenemann, M. (2025). Finding the sweet spot: Spatial resolution and feature set effects on mapping mesquite-encroached rangelands. *Southwest Division of the AAG (SWAAG) Annual Meeting*, Las Cruces, NM.

Hoque, A., & Buenemann, M. (2025). Spatial resolution effects on mapping mesquite-encroached rangelands. *Graduate Research and Arts Symposium (GRAS)*, New Mexico State University, Las Cruces, NM. First Prize, Advanced Research, Data Science.

Hoque, A. (2024). Mapping woody plant encroachment with a multi-sensor approach. *AAG Annual Meeting*, Honolulu, HI.

Hoque, A., & Buenemann, M. (2023). Monitoring changes of rangeland vegetation with remote sensing. *AAG Annual Meeting*, Denver, CO.

Buenemann, M., Akrasi, O., & **Hoque, A.** (2023). Use of multi-temporal optical, radar, and fused optical-radar images for mapping land cover changes in north-central Panama. *AAG Annual Meeting*, Denver, CO.

Hoque, A. (2023). Mapping woody plant encroachment with remote sensing. *Data Science Symposium*, New Mexico State University, Las Cruces, NM.

Poster Presentations

Hoque, A., & Buenemann, M. (2025). Comparing fractional cover mapping of mesquite using machine learning approaches. *American Geophysical Union (AGU) Fall Meeting*, New Orleans, LA.

Hoque, A., Ransom, A., Buenemann, M., & Reichenborn, M. (2022). Use of remote sensing for monitoring rangeland vegetation changes following herbicide treatment. *SWAAG Annual Meeting*, Arkansas.

HONORS, AWARDS, AND FUNDING

Awards

2025	First Prize, Advanced Research (Data Science), Graduate Research and Arts Symposium, NMSU
2024	Best Teaching Assistant Award, Department of Geography and Environmental Studies, NMSU
2024	Jesse Rossbach Scholarship, New Mexico Geographic Information Council (NMGIC)
2019	National Science and Technology (NST) Award for Research Proposal, Government of Bangladesh

Fellowships and Scholarships

2022 – 2026	Graduate Success Scholarship, NMSU (awarded annually)
2023	SWAAG Student Summer Research Scholarship
2021	Erasmus Mobility Exchange Fund for research internship at DLR, University of Twente
2020	Orange Knowledge Programme (OKP) Scholarship, Government of the Netherlands (Nuffic)
2020	ITC Excellence Scholarship, University of Twente

Travel Grants

2023, 2025	College of Arts and Sciences Graduate Student Travel Grant, NMSU
2023, 2024	Department of Geography and Environmental Studies Travel Grant, NMSU
2023	AAG and Gamma Theta Upsilon (GTU) Travel Grant, American Association of Geographers

TEACHING EXPERIENCE

Instructor of Record, Maps and GIScience (undergraduate) Spring 2026
Department of Geography and Environmental Studies, NMSU

- Designed and taught an introductory GIS course, including weekly labs in ArcGIS Online and ArcGIS Pro.

Graduate Teaching Assistant 2023 – Present

Department of Geography and Environmental Studies, NMSU

- Courses: Programming for GIS, Advanced Remote Sensing, and Cartography and GIS, among others.
- Developed a lecture and lab exercise on ArcGIS Online Dashboards for a land cover analysis course; led lab troubleshooting and grading.

Instructor of Record and Lab Instructor 2019 – 2020

Department of Geography and Environmental Studies, University of Chittagong, Bangladesh

- Taught Remote Sensing and Satellite Imagery (instructor of record) and Geographic Information Systems (laboratory).

MENTORING

- Supervised an undergraduate independent study on GIS fundamentals and applied spatial analysis, NMSU (Spring 2026).

TECHNICAL SKILLS

Programming	Python (primary), R, JavaScript (Earth Engine API), Bash
Machine learning	TensorFlow/Keras, scikit-learn; CNN, U-Net, attention-based fusion networks, recurrent networks (GRU), Random Forest, SVM
Remote sensing	Sentinel-1 SAR, Sentinel-2, Landsat, MODIS, SMAP, NAIP, PlanetScope, EMIT, UAS hyperspectral and multispectral imagery
Software	Google Earth Engine, ENVI, SNAP, eCognition, TIMESAT, FRAGSTATS, ArcGIS Pro, ArcGIS Online, QGIS
Computing	High-performance computing with GPU training, Git and GitHub, Jupyter
Field methods	UAS flight planning and operation, field spectroradiometry, vegetation cover sampling

CERTIFICATIONS

2023	FAA Part 107 Remote Pilot Certificate, small unmanned aircraft systems
2024	Python Scripting for Geoprocessing Workflows; Creating Python Scripts for Raster Analysis (Esri)

SERVICE AND LEADERSHIP

2025 – Present	Vice President, Bangladesh Student Association, NMSU
2024 – 2025	President, Graduate Geography and Environmental Studies Student Organization (GGESO), NMSU
2022 – Present	Secretary, Gamma Theta Upsilon International Geographical Honor Society, Kappa Xi Chapter, NMSU
2016 – 2020	Co-founder, National Environmental Awareness and Restoration (NEAR); organized workshops and poster competitions
2016 – 2018	Organizer, GIS Day, University of Chittagong
2018 – 2020	Founding Member, Chittagong University Scientific Society

Professional Memberships

American Association of Geographers (2022 – Present); American Geophysical Union (2023 – Present)

REFERENCES

Available upon request.